

Задание часть 3. Встраиваемые системы

Задание 17 - по ядру Linux

Сделал: Коваленко Дмитрий

- 1) Сначала настроил grub, чтобы при запуске появлялось меню

```
GNU nano 8.7.1
# If you change this file or any /etc/default/grub.d/*.cfg file,
# run 'update-grub' afterwards to update /boot/grub/grub.cfg.
# For full documentation of the options in these files, see:
#   info -f grub -n 'Simple configuration'

GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=menu
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`( . /etc/os-release && echo ${NAME} )`
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
GRUB_CMDLINE_LINUX=""

# If your computer has multiple operating systems installed, then you
# probably want to run os-prober. However, if your computer is a host
# for guest OSes installed via LVM or raw disk devices, running
# os-prober can cause damage to those guest OSes as it mounts
# filesystems to look for things.
#GRUB_DISABLE_OS_PROBER=false
```

- 2) Затем клонировал, зашел в папку linux и запустил make defconfig

```
komal@komal-VirtualBox:~$
komal@komal-VirtualBox:~$ git clone https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git
```

```

komal@komal-VirtualBox:~/linux$ make defconfig
YACC      scripts/kconfig/parser.tab.[ch]
HOSTCC    scripts/kconfig/lexer.lex.o
HOSTCC    scripts/kconfig/menu.o
HOSTCC    scripts/kconfig/parser.tab.o
HOSTCC    scripts/kconfig/preprocess.o
HOSTCC    scripts/kconfig/symbol.o
HOSTCC    scripts/kconfig/util.o
HOSTLD    scripts/kconfig/conf
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
komal@komal-VirtualBox:~/linux$

```

- 3) через make menuconfig настроил будущую сборку ядра так, чтобы хотя запускалось нормально

```

< > DRM Support for bochs disp vga interface
< > Cirrus driver for QEMU emulated device
< > GM12U320 driver for USB projectors
< > DisplayLink
[*] Virtual Box Graphics Card
< > Virtual GEM provider
< * > Virtio GPU driver
[*] Virtio GPU driver modesetting support
Virtual KMS (EXPERIMENTAL)

```

```

--- Mice
< * > PS/2 mouse
[ ] Elantech PS/2 protocol extension
[ ] Sentelic Finger Sensing Pad PS/2 protocol extension
[ ] eGalax TouchKit PS/2 protocol extension
[*] Virtual mouse (vmmouse)
< > Serial mouse
< > Apple USB Touchpad support
< > Apple USB BCM5974 Multitouch trackpad support
< > Cypress APA I2C Trackpad support
< > ELAN I2C Touchpad support
< > DEC VSXXX-AA/GA mouse and VSXXX-AB tablet
< > Synaptics I2C Touchpad support
< > Synaptics USB device support

```

```

--- Virtualization drivers
< * > Virtual Machine Generation ID driver (NEW)
[*] Virtual Box Guest integration support
< > Nitro Enclaves Support (NEW)
< > EFI secret area securityfs support (NEW)

```

```

    ARM System Control and Management Interface Protocol ----
< > BIOS Enhanced Disk Drive calls determine boot disk
[*] Export DMI identification via sysfs to userspace
< > DMI table support in sysfs
< > iSCSI Boot Firmware Table Attributes module
< > QEMU fw_cfg device support in sysfs
[*] Mark VGA/VBE/EFI FB as generic system framebuffer
[ ] Google Firmware Drivers ----
    EFI (Extensible Firmware Interface) Support --->
    Qualcomm firmware drivers ----
    Tegra firmware driver ----

```

```

--- Support for frame buffer device drivers
< > Cirrus Logic support
< > Permedia2 support
< > CyberPro 2000/2010/5000 support
< > Arc Monochrome LCD board support
[ ] Asilant (Chips) 69000 display support
[ ] IMS Twin Turbo display support
< > VGA 16-color graphics support
< > Userspace VESA VGA graphics support
[*] VESA VGA graphics support
[*] EFI-based Framebuffer Support
< > N411 Apollo/Hecuba devkit support
< > Hercules mono graphics support

```

```

(80) Initial number of console screen columns
(25) Initial number of console screen rows
[*] Framebuffer Console support
[ ] Enable legacy fbcon hardware acceleration code (NEW)
[ ] Map the console to the primary display device (NEW)
[ ] Framebuffer Console Rotation (NEW)
[ ] Framebuffer Console Deferred Takeover (NEW)

```

```

[*] Enable legacy fbdev support for your modesetting driver
(100) Overallocation of the fbdev buffer (NEW)
[ ] Print the kernel boot message on the screen
    Default DRM Client (fbdev) --->

```

- 4) Далее прописываем, чтобы началась сборка ядра (`make bzImage && make modules && sudo make modules_install && sudo make install`) и после сборки навеской обновил grub, чтобы мое новосозданное ядро отобразилось в меню (`sudo update-grub`) ну и перезапустил систему (`sudo reboot`)

